HOTS

Both PCR and Cloning allow for the production of many copies of a DNA sequence. What are the advantages of using PCR instead of cloning to amplify a DNA template?

What advantages are there to using a DNA polymerase for PCR that has proofreading activity?

12.11.3 Molecular Markers – RFLP, RAPD

A molecular marker is a DNA sequence in the genome which can be located and identified. As a result of genetic alterations (mutations, insertions, deletions), the base composition at a particular location of the genome may be different in different cells. These differences, collectively called as polymorphisms can be mapped and identified. The DNA markers are highly useful for genetic mapping of genomes. RFLPS (Restriction Fragment Length Polymorphisms), VNTRs (mini satellites or Variable Number Tandem Repeats), STRs (Microsatellites or Simple Random Repeats), SNPs (Single Nucleotide Polymorphisms) are types of DNA sequences (stretch of DNA) which can be used as markers. These markers are used in disease diagnosis and DNA fingerprinting.

Restriction fragment length polymorphism (RFLP) was the very first technology employed for the detection of polymorphism, based on the DNA sequence differences. RFLP is mainly based on the altered restriction enzyme sites, as a result of mutations and recombinations of genomic DNA. An outline of the RFLP analysis is given

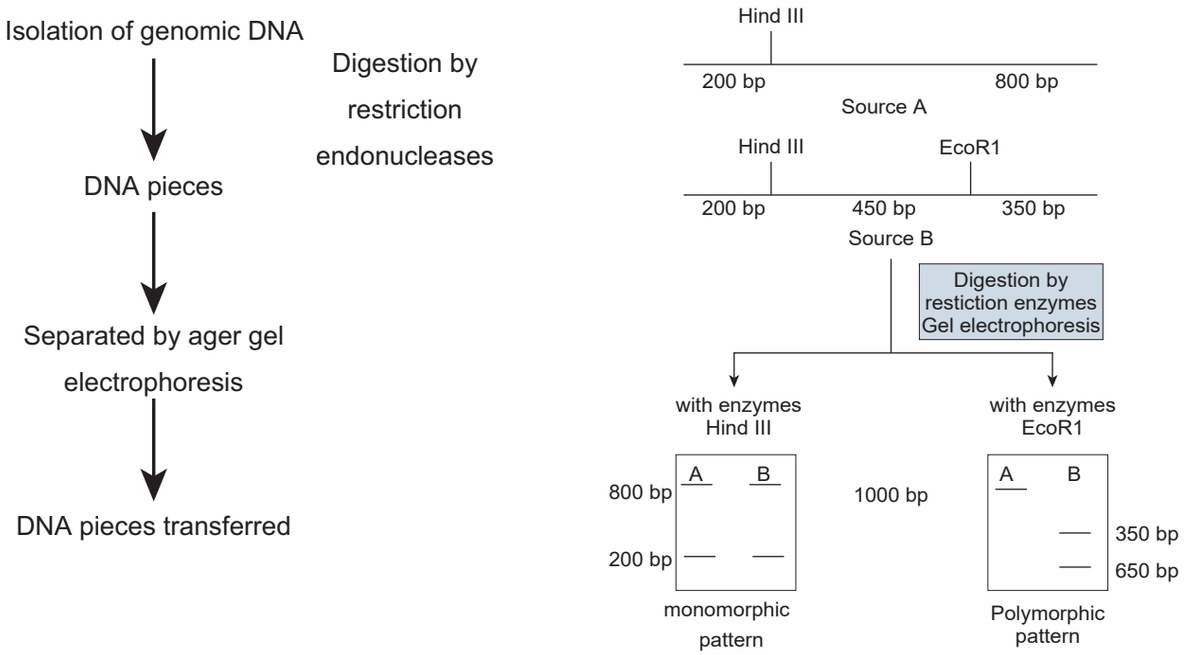


Figure 12.34: (a) An outline of RFLP analysis as a molecular marker **(b)** A schematic representation of restriction fragment length polymorphism (RFLP) analysis as a molecular marker

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